

```
global_data={
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "load_losses_watts",
  "attribute_4": "unit_volume_liters",
  "attribute_5": "service_life_years",

  "direction_attribute_1": "negative",
  "direction_attribute_2": "positive",
  "direction_attribute_3": "negative",
  "direction_attribute_4": "negative",
  "direction_attribute_5": "positive",

  "attribute_1_min": 2000,
  "attribute_1_max": 50000,
  "attribute_2_min": 0,
  "attribute_2_max": 1,
  "attribute_3_min": 1000,
  "attribute_3_max": 7000,
  "attribute_4_min": 300,
  "attribute_4_max": 2000,
  "attribute_5_min": 10,
  "attribute_5_max": 50,

  "copper_price_per_ton": 9500,
  "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.5,
  "aluminum_price_per_kg": 2.5
}
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regional_data=[
  {
    "region": "United States",
    "copper_product_market_share": 0.45,
    "aluminum_product_market_share": 0.55,
    "weight_attribute_1": 0.225,
    "weight_attribute_2": 0.100,
    "weight_attribute_3": 0.360,
    "weight_attribute_4": 0.135,
    "weight_attribute_5": 0.180
  },
  {
    "region": "China",
    "copper_product_market_share": 0.60,
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    "aluminum_product_market_share": 0.40,
    "weight_attribute_1": 0.360,
    "weight_attribute_2": 0.100,
    "weight_attribute_3": 0.270,
    "weight_attribute_4": 0.135,
    "weight_attribute_5": 0.135
  },
  {
    "region": "India",
    "copper_product_market_share": 0.45,
    "aluminum_product_market_share": 0.55,
    "weight_attribute_1": 0.405,
    "weight_attribute_2": 0.100,
    "weight_attribute_3": 0.225,
    "weight_attribute_4": 0.090,
    "weight_attribute_5": 0.180
  },
  {
    "region": "Europe",
    "copper_product_market_share": 0.65,
    "aluminum_product_market_share": 0.35,
    "weight_attribute_1": 0.135,
    "weight_attribute_2": 0.100,
    "weight_attribute_3": 0.405,
    "weight_attribute_4": 0.180,
    "weight_attribute_5": 0.180
  }
]

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product_data=[
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "250 kVA ONAN 3-Phase Distribution Transformer (Cu-Wound)",
    "copper_kg": 180,
    "aluminum_kg": 0,
    "total_mass_kg": 900,
    "non_material_cost_per_unit": 13290,
    "attribute_2_value": 1,
    "attribute_3_value": 2700,
    "attribute_4_value": 850,
    "attribute_5_value": 35
  },
  {

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"region": "United States",
"dominant material": "al",
"product": "250 kVA ONAN 3-Phase Distribution Transformer (Al-Wound)",
"copper_kg": 0,
"aluminum_kg": 300,
"total_mass_kg": 1100,
"non_material_cost_per_unit": 12250,
"attribute_2_value": 0,
"attribute_3_value": 3250,
"attribute_4_value": 1100,
"attribute_5_value": 30
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "250 kVA ONAN 3-Phase Distribution Transformer (Cu-Wound)",
  "copper_kg": 180,
  "aluminum_kg": 0,
  "total_mass_kg": 900,
  "non_material_cost_per_unit": 7290,
  "attribute_2_value": 1,
  "attribute_3_value": 2700,
  "attribute_4_value": 850,
  "attribute_5_value": 35
},
{
  "region": "China",
  "dominant material": "al",
  "product": "250 kVA ONAN 3-Phase Distribution Transformer (Al-Wound)",
  "copper_kg": 0,
  "aluminum_kg": 300,
  "total_mass_kg": 1100,
  "non_material_cost_per_unit": 6750,
  "attribute_2_value": 0,
  "attribute_3_value": 3250,
  "attribute_4_value": 1100,
  "attribute_5_value": 30
},
{
  "region": "India",
  "dominant material": "cu",
  "product": "250 kVA ONAN 3-Phase Distribution Transformer (Cu-Wound)",
  "copper_kg": 180,
  "aluminum_kg": 0,
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"total_mass_kg": 900,
"non_material_cost_per_unit": 6290,
"attribute_2_value": 1,
"attribute_3_value": 2700,
"attribute_4_value": 850,
"attribute_5_value": 35
},
{
  "region": "India",
  "dominant material": "al",
  "product": "250 kVA ONAN 3-Phase Distribution Transformer (Al-Wound)",
  "copper_kg": 0,
  "aluminum_kg": 300,
  "total_mass_kg": 1100,
  "non_material_cost_per_unit": 5750,
  "attribute_2_value": 0,
  "attribute_3_value": 3250,
  "attribute_4_value": 1100,
  "attribute_5_value": 30
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "250 kVA ONAN 3-Phase Distribution Transformer (Cu-Wound)",
  "copper_kg": 180,
  "aluminum_kg": 0,
  "total_mass_kg": 900,
  "non_material_cost_per_unit": 13290,
  "attribute_2_value": 1,
  "attribute_3_value": 2700,
  "attribute_4_value": 850,
  "attribute_5_value": 35
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "250 kVA ONAN 3-Phase Distribution Transformer (Al-Wound)",
  "copper_kg": 0,
  "aluminum_kg": 300,
  "total_mass_kg": 1100,
  "non_material_cost_per_unit": 11750,
  "attribute_2_value": 0,
  "attribute_3_value": 3250,
  "attribute_4_value": 1100,
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    "attribute_5_value": 30
  }
]
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